AI Assistant lab-21.3

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TASK 1 – AI-Assisted Expense Tracker

Initial prompt:

Create a simple Python expense tracker to store daily expenses with category and amount.

Initial code:

```
expenses = []

while True:
    name = input("Enter expense name: ")
    amount = float(input("Enter amount: "))
    category = input("Enter category: ")

    expenses.append((name, amount, category))

    choice = input("Add more? (y/n): ")
    if choice == 'n':
        break

print("Expenses:", expenses)
```

Output:

```
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ /usr/bin/python3
/home/shashidhar/Downloads/aiassitantcoding-20260312T084629Z-3-001/aiassitantcoding/65.py
Enter expense name: food
Enter amount: 100
Enter category: grocieries
Add more? (y/n): y
Enter expense name: transport
Enter amount: 50
Enter category: travel
Add more? (y/n): n
Expenses: [('food', 100.0, 'grocieries'), ('transport', 50.0, 'travel')]
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$
```

Improvement prompt:

Improve the expense tracker with monthly summary, input validation, and functions.

Improved code

```
def add_expense(expenses):
```

```python
    name = input("Enter expense name: ")

    while True:
        try:
            amount = float(input("Enter amount: "))
            break
        except:
            print("Invalid amount")

    category = input("Enter category: ")
    expenses.append({"name": name, "amount": amount, "category": category})


def show_summary(expenses):
    total = sum(e["amount"] for e in expenses)
    print("Total Monthly Expense:", total)


expenses = []

while True:
    add_expense(expenses)
    choice = input("Add more? (y/n): ")
    if choice == 'n':
        break

show_summary(expenses)
```

Output:

```
PROBLEMS 6   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   ...   Python

shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ /usr/bin/python3
/home/shashidhar/Downloads/aiassitantcoding-20260312T084629Z-3-001/aiassitantcoding/65.py
Enter expense name: food
Enter amount: 100
Enter category: grocieries
Add more? (y/n): y
Enter expense name: transport
Enter amount: 50
Enter category: travel
Add more? (y/n): n
Total Monthly Expense: 150.0
```

Justification

- Functions → modular design

- Validation → avoids runtime errors
- Summary → real-world usability

TASK 2 – Git Version Control (Student Grade Calculator)

Initialize Repo,Initial Code,Create Branch

```
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$  cd /home/shashidhar/Downloads/grade-calculator
 && git add . && git commit -m "Initial grade calculator with average"
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'shashidhar@shashidhar-750XGK.(none)')
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$  git config user.name "shashidhar" && git config user.email "shashidhar@example.com" && git add .
mit -m "Initial grade calculator with average"
[master (root-commit) 15776bb] Initial grade calculator with average
 1 file changed, 3 insertions(+)
 create mode 100644 grade.py
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$  cd /home/shashidhar/Downloads/grade-calculator && git checkout -b feature-ai-grade && cat > grad
'
> marks = list(map(int, input("Enter marks: ").split()))
>
> total = sum(marks)
> avg = total / len(marks)
> percentage = (total / (len(marks)*100)) * 100
>
> if avg >= 90:
>     grade = 'A'
> elif avg >= 75:
>     grade = 'B'
> elif avg >= 50:
>     grade = 'C'
> else:
>     grade = 'F'
>
> print("Average:", avg)
> print("Percentage:", percentage)
> print("Grade:", grade)
> EOF
Switched to a new branch 'feature-ai-grade'
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$
```

Ai prompt

Add grade classification and percentage calculation to the program.
Updated

```
marks = list(map(int, input("Enter marks: ").split()))

total = sum(marks)
avg = total / len(marks)
percentage = (total / (len(marks)*100)) * 100

if avg >= 90:
   grade = 'A'
elif avg >= 75:
   grade = 'B'
```

```
elif avg >= 50:
    grade = 'C'
else:
    grade = 'F'

print("Average:", avg)
print("Percentage:", percentage)
print("Grade:", grade)
```

Commit + Merge

```
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$ git add . && git commit -m "Added grade classification and percentage calculation"
 1 file changed, 16 insertions(+), 1 deletion(-)
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$  git status --short && git log --oneline --decorate --graph --all --max-count=5 && printf '\n---
-\n' && sed -n '1,120p' grade.py
* 649c76d (HEAD -> feature-ai-grade) Added grade classification and percentage calculation
* 15776bb (master) Initial grade calculator with average

--- grade.py ---
marks = list(map(int, input("Enter marks: ").split()))

total = sum(marks)
avg = total / len(marks)
percentage = (total / (len(marks)*100)) * 100

if avg >= 90:
    grade = 'A'
elif avg >= 75:
    grade = 'B'
elif avg >= 50:
    grade = 'C'
else:
    grade = 'F'

print("Average:", avg)
print("Percentage:", percentage)
print("Grade:", grade)
shashidhar@shashidhar-750XGK:~/Downloads/grade-calculator$  git branch -m master main && git checkout main && git merge feature-ai-grade && git log --onelin
e --graph --all --max-count=5
Switched to branch 'main'
Updating 15776bb..649c76d
Fast-forward
 grade.py | 17 ++++++++++++++++-
 1 file changed, 16 insertions(+), 1 deletion(-)
* 649c76d (HEAD -> main, feature-ai-grade) Added grade classification and percentage calculation
* 15776bb Initial grade calculator with average
```

Justification

- Branching → safe feature development
- Merge → integration workflow
- AI → faster feature addition

TASK 3 – AI Commit Messages (Password Validation)

Setup code

```python
import re


password = input("Enter password: ")


if len(password) < 8:
    print("Weak password")
elif not re.search(r'[!@#$%^&*]', password):
    print("Add special character")
else:
    print("Strong password")
```

Prompt

Generate commit message for adding password validation with length and special character rules.

AI Commit Message

```
fatal: not a git repository (or any of the parent directories): .git
shashidhar@shashidhar-750XGK:~/Downloads/aiassitantcoding-20260312T084629Z-3-001$ git commit -m "feat: add password validation with minimum length and special char
acter requirement"
```

Human Version

Added strong password validation rules

Justification

- AI → more structured (uses “feat:” convention)
- Human → less descriptive

TASK 4 – Quiz Management System (Pair Programming)

Student A Code

```
questions = {
   "Capital of India?": "Delhi",
   "2+2?": "4"
}

for q in questions:
   ans = input(q + " ")
   if ans == questions[q]:
       print("Correct")
   else:
       print("Wrong")
```

AI Prompt (Student B)

Add score calculation and input validation to quiz program.

Final code

```
questions = {
   "Capital of India?": "Delhi",
   "2+2?": "4"
}

score = 0

for q in questions:
   ans = input(q + " ").strip()

   if ans.lower() == questions[q].lower():
       print("Correct")
       score += 1
   else:
       print("Wrong")

print("Final Score:", score)
```

Git Workflow

```
shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git checkout -b feature-score
Switched to a new branch 'feature-score'

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git add .

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git commit -m "Added score calculation"
[feature-score 8ab91cd] Added score calculation
 1 file changed, 12 insertions(+), 1 deletion(-)

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git checkout main
Switched to branch 'main'

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git merge feature-score
Updating 2f4a7d1..8ab91cd
Fast-forward
 quiz.py | 12 ++++++++++-
 1 file changed, 11 insertions(+), 1 deletion(-)

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ python quiz.py
Capital of India? Delhi
Correct
2+2? 5
Wrong
Final Score: 1

shashidhar@shashidhar-750XGK:~/Downloads/quiz-management$ git log --oneline --decorate --graph --all --max-count=5
* 8ab91cd (HEAD -> main, feature-score) Added score calculation
* 2f4a7d1 Initial quiz management system
```

Output:

```
PROBLEMS 79   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS   GITLENS   SPELL CHECKER 69   SONARQUBE 4   Python - Downloads

shashidhar@shashidhar-750XGK:~/Downloads$ /usr/bin/python3 /home/shashidhar/Downloads/aiassitantcoding-20260312T084629Z-3-001/aiassitantcoding/65.py
Capital of India? y
Wrong
2+2? 4
Correct
Final Score: 1
shashidhar@shashidhar-750XGK:~/Downloads$
```

Justification

- Score → measurable output
- Validation → better UX
- Collaboration → real Git workflow